

ZHEZQAZGHAN, KazakhstanWMO: **356710**Lat: **47.800N**Lon: **67.717E**Elev: **1135**StdP: **14.10**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.4	-16.5	-27.6	1.2	-20.9	-22.8	1.6	-16.2	27.4	19.1	24.2	17.3	4.6	340	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	96.6	64.8	93.4	63.7	90.3	62.7	66.7	88.9	65.5	87.3	64.2	85.7	10.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	81.1	71.4	58.1	75.2	71.1	56.0	69.5	71.5	31.9	88.7	30.9	87.3	30.0	85.6	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	22.6	19.6	-26.1	102.0	5.6	2.8	-30.1	104.1	-33.3	105.7	-36.5	107.3	-40.5	109.4		
(5)			-26.8	69.3	5.9	2.3	-31.0	70.9	-34.5	72.3	-37.8	73.6	-42.0	75.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	6.9	9.3	27.3	48.9	60.9	71.7	74.3	72.1	59.5	44.4	27.6	13.1
	DBStd	26.28	12.58	12.63	13.07	9.77	7.74	6.87	6.28	7.33	8.58	9.24	10.97	12.38
	HDD50	5381	1335	1139	709	134	9	0	0	15	221	674	1145	
	HDD65	8813	1800	1559	1169	484	172	20	5	24	208	639	1123	1610
	CDD50	2890	0	0	5	102	348	650	754	684	299	47	1	0
	CDD65	850	0	0	0	3	45	220	295	243	43	1	0	0
	CDH74	11624	0	0	2	70	752	3034	3831	3234	675	26	0	0
	CDH80	5304	0	0	0	8	212	1407	1863	1566	246	2	0	0
Wind	WSAvg	8.7	8.0	8.2	9.9	9.9	9.9	9.3	9.2	8.8	8.0	8.0	7.5	8.2
Precipitation	PrecAvg	7.5	0.7	0.5	0.7	0.7	0.7	0.6	0.7	0.5	0.4	0.7	0.6	0.9
	PrecMax	18.4	1.8	1.1	1.9	2.3	4.1	3.4	1.8	3.0	2.0	1.7	2.0	11.8
	PrecMin	3.6	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	3.1	0.4	0.3	0.4	0.6	0.9	0.7	0.6	0.6	0.6	0.5	0.4	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	36.6	66.2	80.6	89.7	98.6	100.8	99.5	92.9	77.7	56.1	37.1
		MCWB	33.4	34.4	51.7	57.4	60.8	64.2	66.5	65.9	62.1	53.9	49.6	34.0
	2%	DB	32.4	33.3	56.7	75.5	85.4	94.5	96.7	95.4	86.6	71.4	50.3	33.8
		MCWB	31.4	31.6	47.0	54.7	59.9	63.8	65.0	64.3	59.3	51.1	44.1	32.0
	5%	DB	29.1	31.0	50.0	71.0	81.9	91.4	93.2	92.3	81.6	66.3	45.6	31.8
		MCWB	28.0	29.5	43.2	52.4	58.4	63.0	63.9	63.4	57.2	49.0	40.6	30.4
	10%	DB	25.6	28.2	43.8	66.5	78.1	88.0	89.6	89.1	76.8	60.9	41.4	29.3
		MCWB	24.6	26.7	39.3	50.4	56.8	62.1	62.9	62.4	55.6	47.3	37.8	27.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	34.8	52.2	59.3	64.0	68.6	68.4	68.5	63.2	55.7	51.7	34.7
		MCDB	34.3	36.4	65.0	75.7	80.8	88.4	90.5	93.9	89.7	72.4	55.4	35.6
	2%	WB	31.5	31.9	47.7	56.5	61.6	66.2	66.8	65.9	60.3	52.7	45.1	32.2
		MCDB	32.4	33.1	55.4	72.5	80.1	87.5	88.6	90.3	83.1	67.6	49.0	33.2
	5%	WB	28.2	29.6	43.5	53.9	59.7	64.8	65.7	64.5	58.1	50.3	41.0	30.2
		MCDB	29.1	30.9	49.5	68.0	77.9	85.8	86.9	88.8	78.6	63.7	44.6	31.7
	10%	WB	24.5	26.7	39.4	51.7	58.0	63.3	64.5	63.0	56.2	47.9	38.0	27.8
		MCDB	25.5	28.1	44.1	64.4	75.4	83.7	85.1	85.8	75.2	59.7	41.4	29.2
Mean Daily Temperature Range	5% DB	MDBR	15.4	17.2	16.8	23.0	25.4	25.5	24.6	25.6	26.2	22.7	15.6	15.0
		MCDBR	10.6	13.1	23.7	29.4	30.4	29.8	29.2	30.6	32.3	31.1	20.6	13.0
		MCWBR	10.4	12.6	15.0	13.7	12.6	10.8	10.3	11.5	14.1	16.7	15.3	11.8
	5% WB	MCDBR	10.7	13.1	22.8	26.7	27.2	26.0	26.0	27.8	29.5	28.1	18.0	11.8
		MCWBR	10.6	12.7	15.2	13.8	12.4	10.3	9.5	10.7	13.8	16.2	14.1	11.1
Clear-Sky Solar Irradiance	taub	0.306	0.342	0.394	0.424	0.434	0.426	0.427	0.397	0.355	0.351	0.337	0.313	
	taud	2.079	1.987	2.046	2.078	2.108	2.199	2.218	2.283	2.353	2.275	2.201	2.128	
	Ebn at Noon	232	248	256	261	264	266	264	267	269	247	220	213	
	Edh at Noon	29	40	45	48	48	44	43	38	32	30	26	25	
All-Sky Solar Radiation	RadAvg	490	850	1225	1655	2099	2223	2119	1929	1496	886	461	352	
	RadStd	82	89	120	151	140	131	105	111	56	79	43	59	